EDITOR FS

MASTER ARCHITECTURE & CODE MAP

Complete Current-Codebase Architecture & Developer Reference Manual

Project: Editor FS — Flutter Video Editor
Package: com.example.capcut_video_editor
Architecture: Flutter / Native Android Custom Hardware Engine

Documentation Scope: Current Repository Implementation
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1. Master System Architecture

Editor FS uses a strict layered boundary between the Flutter user experience layer and the native Android hardware engine. The application contains **zero third-party media dependencies** (no `video_player`, no `audioplayers`, no `flutter_ffmpeg`). All media decoding, texture rendering, audio mixing, and MP4 hardware encoding are executed on native Android hardware via custom platform channels.

FLUTTER PRESENTATION LAYER
EditorScreen • VideoPreviewSection • TimelineWidget • BottomToolbar • Modal Tool Sheets

STATE MANAGEMENT & DOMAIN (EditorViewModel)
Timeline State • Authoritative Playhead Clock • Undo/Redo • MediaAsset • Project Drafts

PLATFORM CHANNEL INTERFACES (Services)
VideoPlaybackService • AudioPlayerService • FilePickerService • TtsService

NATIVE ANDROID HARDWARE ENGINES (Kotlin)
MainActivity (16ms Ticker) • VideoPlaybackEngine • VideoExportEngine • AudioEngine • MediaAssetEngine

Data Flow & Ownership

Flutter Side: User actions mutate `EditorViewModel`, which dispatches immutable project state updates and sends imperative commands to native channels.
Native Side: Owns OS media handles (MediaPlayer, MediaCodec, EGLSurface, AudioTrack). Streams position and completion events back to Flutter.

Hardware Boundaries

Native/Flutter Boundary: BinaryMessenger passing JSON/primitives across 4 MethodChannels and 2 EventChannels.
Media Processing Boundary: Video frames stay on Android GPU surfaces; Flutter only receives a surface `textureId` for direct hardware rendering.

2. Complete Project Directory Map

```
capcut_video_editor/
├── pubspec.yaml                           # Zero media deps (cupertino_icons, uuid, flutter_lints)
├── android/app/src/main/
│   ├── AndroidManifest.xml               # Storage, Audio & Camera permissions, Hardware acceleration
│   └── kotlin/com/example/capcut_video_editor/
│       ├── MainActivity.kt               # Channel routers & 16ms authoritative clock ticker
│       ├── VideoPlaybackEngine.kt        # MediaPlayer & Flutter TextureRegistry bridge
│       ├── VideoExportEngine.kt          # Hardware MediaCodec, EGL14/GLES20, 12 transitions, Muxer
│       ├── AudioEngine.kt                # Multi-track audio player & voiceover recorder
│       └── MediaAssetEngine.kt           # Metadata extraction, thumbnail gen, fast audio demux
├── lib/
│   ├── main.dart                         # Application entry point, multi-provider setup, dark theme
│   ├── models/                           # Domain Data Entities
│   │   ├── media_asset.dart              # Visual asset model (video, photo, sticker, b-roll)
│   │   ├── audio_track.dart              # Audio entity (BGM, sound effects, voiceover, volume, trim)
│   │   ├── text_overlay.dart             # Text overlay model (style, animation, font, coordinates)
│   │   ├── sticker_overlay.dart          # Sticker entity (emoji/image, scale, rotation, coordinates)
│   │   ├── transition_effect.dart        # 12 GPU transition types enum & duration bounds
│   │   ├── video_effect.dart             # Visual filter & shader adjustment definitions
│   │   ├── project.dart                  # Root project draft model serializable to JSON
│   │   ├── timeline_track.dart           # Multi-track timeline track hierarchy definitions
│   │   └── export_preset.dart            # Resolution, framerate & bitrate presets
│   ├── viewmodels/                       # State Management
│   │   └── editor_view_model.dart        # Central editor state: clock, tracks, history, undo/redo
│   ├── presentation/
│   │   ├── screens/
│   │   │   ├── editor_screen.dart        # Primary editing workstation UI viewport
│   │   │   ├── home_screen.dart          # Project library, draft manager & launcher
│   │   │   └── export_screen.dart        # Export configuration, progress modal & gallery launcher
│   │   └── widgets/                      # Interactive UI Components
│   │       ├── video_preview_section.dart # Texture renderer & real-time 12-transition canvas
│   │       ├── timeline_widget.dart       # Multi-track timeline, playhead scrubber, split/trim handles
│   │       ├── bottom_toolbar.dart        # Primary action dock (Edit, Audio, Text, Stickers, Canvas)
│   │       ├── audio_bottom_sheet.dart    # Audio management (BGM, SFX library, voice recorder, TTS)
│   │       ├── text_bottom_sheet.dart     # Typography, styling, color palette & entry modal
│   │       ├── speed_bottom_sheet.dart    # Speed ramp (0.1x to 100x) & pitch preservation selector
│   │       ├── transition_selector.dart   # Visual 12-transition picker with duration slider
│   │       ├── filter_selector.dart       # Color grading & LUT preset picker
│   │       ├── effect_selector.dart       # Visual effects & shader overlay selector
│   │       ├── ratio_selector.dart        # Canvas aspect ratio picker (9:16, 16:9, 1:1, 4:5)
│   │       ├── sticker_bottom_sheet.dart  # Emoji & sticker picker drawer
│   │       └── volume_bottom_sheet.dart   # Track audio gain & fade-in/fade-out sliders
│   └── services/                         # Core Device & Platform Services
│       ├── video_playback_service.dart   # Facade for video playback MethodChannel & EventChannel
│       ├── audio_player_service.dart     # Facade for audio playback MethodChannel & EventChannel
│       ├── file_picker_service.dart      # Facade for native gallery picker & asset extraction
│       ├── tts_service.dart              # Facade for text-to-speech engine
│       ├── project_storage_service.dart  # Draft save/load engine writing JSON files to local disk
│       └── asset_library_service.dart    # Bundled SFX, music tracks, stickers & fonts registry
└── test/                                 # Test Suites (263 Tests Verified)
    ├── unit/
    │   ├── editor_view_model_test.dart   # ViewModel state, clock, play/pause, split/trim, sync
    │   ├── project_storage_service_test.dart# JSON serialization & draft persistence fidelity
    │   ├── transition_effect_test.dart   # Transition model bounds & parameter validation
    │   └── audio_track_test.dart         # Audio track timing, trimming & gain validation
    └── widget/
        ├── editor_screen_test.dart       # UI layout, preview positioning & toolbar interaction
        └── timeline_widget_test.dart     # Scrubber drag gestures, track zooming & playhead
```

3. File-by-File Code Map

Exhaustive index of primary source files with exact classes, functions, layers, and responsibilities.

`lib/viewmodels/editor_view_model.dart` — State Management

Responsibility:	Single source of truth for editor state, playhead position, track collections, undo/redo history, and native sync.
Important Classes:	`EditorViewModel` (extends `ChangeNotifier`)
Key Functions:	`play()` , `pause()` , `seekTo(seconds)` , `splitCurrentClip()` , `trimClip()` , `deleteClip()` , `rippleDelete()` , `addTransition()` , `_handleVideoPositionEvent()` , `saveDraft()` , `exportProject()`
Dependencies:	`VideoPlaybackService` , `AudioPlayerService` , `ProjectStorageService` , Models (`Project` , `MediaAsset`)
Boundaries:	Receives authoritative position stream from native channel; clamps playhead to `totalDuration` on end.

`android/app/src/main/kotlin/.../MainActivity.kt` — Native Host / Routing

Responsibility:	Registers Flutter platform channels, dispatches method calls to native engines, runs 16ms clock ticker.
Important Classes:	`MainActivity` (extends `FlutterActivity`)
Key Functions:	`configureFlutterEngine()` , `startVideoPositionUpdates()` , `stopVideoPositionUpdates()` , `dispatchExport()`
Dependencies:	`VideoPlaybackEngine` , `VideoExportEngine` , `AudioEngine` , `MediaAssetEngine`
Boundaries:	Owns the 16ms `Handler(Looper.getMainLooper())` periodic ticker driving `onPositionUpdate` .

`android/app/src/main/kotlin/.../VideoPlaybackEngine.kt` — Native Video Playback

Responsibility:	Hardware video decoding via Android `MediaPlayer` into Flutter `SurfaceTexture` .
Important Classes:	`VideoPlaybackEngine`
Key Functions:	`initialize(uri, result)` , `play()` , `pause()` , `seekTo(msec)` , `setPlaybackSpeed(speed)` , `setVolume(vol)` , `release()`
Dependencies:	Android `MediaPlayer` , `Surface` , `TextureRegistry.SurfaceTextureEntry`
Boundaries:	Produces native `textureId` passed to Flutter for zero-copy GPU video preview rendering.

`android/app/src/main/kotlin/.../VideoExportEngine.kt` — Hardware Export Engine

Responsibility:	Encodes timeline project to MP4 using hardware `MediaCodec` (H.264/AAC), EGL14/OpenGL, and 12 transitions.
Important Classes:	`VideoExportEngine` , `InputSurface` , `TextureRender`
Key Functions:	`exportVideo(config, onProgress, onComplete, onError)` , `compositeTransition()` , `drainEncoder()` , `setupMuxer()`
Dependencies:	`MediaCodec` , `MediaExtractor` , `MediaMuxer` , `EGL14` , `GLES20` , `MediaStore`
Boundaries:	Enforces 16-pixel dimension alignment for hardware encoder; composites transitions directly on GPU frames.

`lib/presentation/widgets/video_preview_section.dart` — Presentation / Preview

Responsibility:	Renders native video texture, text/sticker overlays, and executes real-time 12-transition canvas math.
Important Classes:	`VideoPreviewSection` , `_VideoPreviewSectionState`
Key Functions:	`_buildVideoTexture()` , `_buildTransitionCanvas()` , `_buildOverlayLayer()`
Dependencies:	`EditorViewModel` , `Texture` widget, Flutter `Transform` , `Opacity` , `ClipRect`
Boundaries:	Calculates transition progress `t = (playhead - start) / dur` to match export pixel for pixel.

`lib/presentation/widgets/timeline_widget.dart` — Presentation / Timeline

Responsibility:	Interactive timeline viewport, multi-track rendering, playhead scrubber drag, and clip trim handles.
Important Classes:	`TimelineWidget` , `_TimelineWidgetState`
Key Functions:	`_buildRuler()` , `_buildVideoTrack()` , `_buildAudioTracks()` , `_handleHorizontalDragUpdate()`
Dependencies:	`EditorViewModel` , `TimelineTrack` , `MediaAsset`
Boundaries:	Translates pixel drag offsets to exact seconds in `EditorViewModel.seekTo()` .

4. Flutter Architecture Map

The Flutter application follows a strictly decoupled Clean Architecture pattern with unidirectional data flow:

Layer	Primary Classes / Files	Dependencies	Data Flow & Purpose
Application Entry	`main.dart`	`ChangeNotifierProvider`, `EditorViewModel`	Initializes Flutter bindings, injects providers, applies global dark editor theme.

Presentation / Screens	`EditorScreen`
	`HomeScreen`
	`ExportScreen`
		`EditorViewModel`, UI Widgets	Renders UI viewports; captures user taps, gestures, and slider inputs; dispatches actions to ViewModel.

Presentation / Widgets	`VideoPreviewSection`
	`TimelineWidget`
	`BottomToolbar`
	Tool Bottom Sheets
		`EditorViewModel`, Flutter Canvas	Interactive controls. Preview computes 12-transition matrices; timeline renders tracks and scrub head.

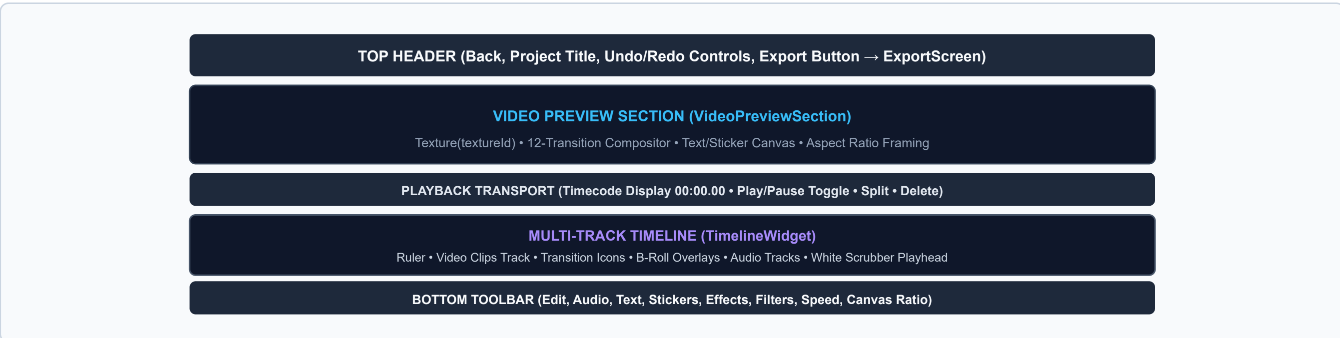
State Management	`EditorViewModel`	Platform Services, Domain Models	Maintains active project state, manages playhead clock, history undo/redo stacks, validates edits.
Domain Entities	`Project`, `MediaAsset`, `AudioTrack`, `TransitionEffect`	Dart standard library, `uuid`	Immutable domain models defining clips, coordinates, volumes, durations, and JSON serialization.

Platform Services	`VideoPlaybackService`
	`AudioPlayerService`
	`FilePickerService`
		Flutter `MethodChannel`, `EventChannel`	Wraps low-level IPC communication to native Android engines into type-safe async Dart methods.

Storage Services	`ProjectStorageService`	`dart:io`, `dart:convert`	Serializes project graphs to JSON files in app-private temporary and persistent storage directories.

5. Editor Screen Architecture

`EditorScreen` serves as the primary IDE workstation. It hosts five major UI zones coordinated through `EditorViewModel`:



6. Timeline Architecture

The timeline represents time along a continuous horizontal axis measured in seconds (double precision).

Timeline Hierarchy Tree

```
Project (id, title, duration, canvasAspectRatio)
 ├── Primary Video Track (List<MediaAsset> placed end-to-end with ripple support)
 │     └── In-Between Adjacent Clips: List<TransitionEffect> (12 types, duration 0.1s - 2.0s)
 ├── Overlay Video Track / PIP (List<MediaAsset> with custom startTime, x/y coords, scale)
 ├── Audio Tracks (List<AudioTrack> with startTime, duration, volume, fadeIn, fadeOut, speed)
 │     ├── Background Music Tracks (BGM)
 │     ├── Sound Effects (SFX)
 │     ├── Extracted Audio (Demuxed from video clips)
 │     └── Voiceover / TTS Audio Clips
 ├── Text Overlays (List<TextOverlay> with startTime, duration, font, style, position, scale)
 └── Sticker Overlays (List<StickerOverlay> with startTime, duration, emoji/image, scale, rotation)
```

Timeline Invariants Enforced by Code

1. **Zero Negative Gaps:** Primary track clips are sequentially adjacent; ripple delete automatically closes gaps.
2. **Transition Adjacency Rule:** A transition can only exist between two contiguous clips `clip[i]` and `clip[i+1]`.
3. **Duration Clamping:** Max transition duration cannot exceed half the duration of either adjoining clip: `dur ≤ min(len(A), len(B)) / 2`.

Playhead Coordinates & Scaling

1. **Pixels Per Second (PPS):** Dynamic scaling based on zoom level (default 50px/sec).
2. **Clamping:** Playhead time is strictly clamped: `0.0 ≤ playhead ≤ totalDuration`.
3. **Scrubbing Mutex:** Manual dragging sets `_isScrubbing = true`, silencing incoming native position ticks until drag ends.

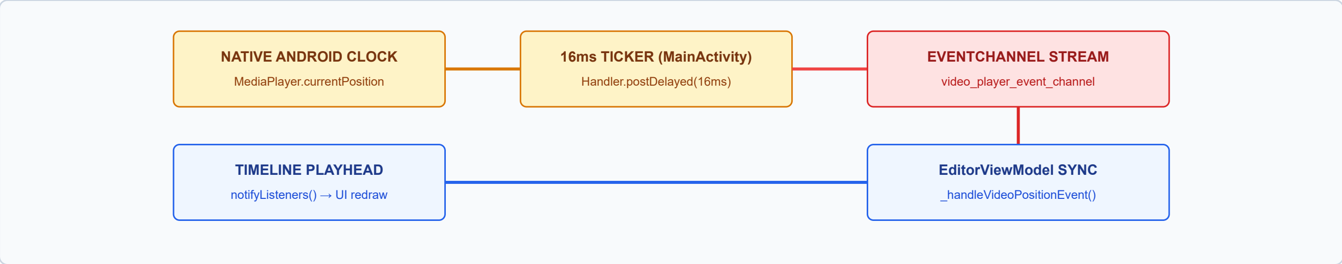
7. MediaAsset Architecture

Every imported visual asset is encapsulated by the immutable `MediaAsset` domain model:

Lifecycle Stage	Code Path / Engine	State Transformation
1. Discovery & Import	`FilePickerService.pickVideo()` → `MediaAssetEngine.kt`	Launches Android SAF / PhotoPicker; extracts content URI and copies to cache path if needed.
2. Metadata Extraction	`MediaAssetEngine.getVideoMetadata()`	Uses Android `MediaMetadataRetriever` to fetch exact width, height, duration, rotation, and fps.
3. Thumbnail Generation	`MediaAssetEngine.getVideoThumbnail()`	Extracts first keyframe as JPEG thumbnail stored in app cache for fast timeline track rendering.
4. Audio Demuxing	`MediaAssetEngine.extractAudioFromVideo()`	Fast container demux via `MediaExtractor` & `MediaMuxer` to standalone `.m4a` without transcoding.
5. Timeline Insertion	`EditorViewModel.addMediaAsset()`	Instantiates `MediaAsset(id, path, duration, type)` and appends to project primary track.
6. Persistence & Deletion	`ProjectStorageService.saveProject()`	Saves relative file paths in JSON draft; asset files remain preserved until explicitly purged.

8. Playback Clock Architecture

A core architectural principle of Editor FS is the **Single Authoritative Playback Clock**. Flutter software timers are completely bypassed during active playback.



Playback Synchronization Invariants

- **Fallback Timer Annihilation:** Any active Flutter software ticker is canceled immediately upon receipt of the first native event.
- **End-of-Playback Clamping:** When native player reports completion or position ≥ `totalDuration`, the ViewModel forces `_playheadPosition = totalDuration`, sets `_isPlaying = false`, stops audio tracks, and stops the ticker.
- **Seek Protection:** When user seeks, native player pauses or completes seek before resume, avoiding stale-position jumps.

9. Audio Subsystem Architecture

The audio subsystem supports concurrent multi-track mixing across 4 distinct track types:

Audio Category	Domain Model	Playback Mechanism	Hardware Export Mixing
Background Music	`AudioTrack(type: bgm)`	Native `MediaPlayer` instance	Decoded to raw PCM, scaled by volume, mixed into master audio bus.
Sound Effects (SFX)	`AudioTrack(type: sfx)`	Native `SoundPool` for zero-latency	Mixed at exact `startTime` timestamp into PCM master buffer.
Extracted Audio	`AudioTrack(type: extracted)`	Native `MediaPlayer` from demuxed M4A	Decoded to PCM; volume envelope and fade applied; mixed into bus.
Voiceover & TTS	`AudioTrack(type: voiceover)`	Recorded via `MediaRecorder` or TTS	Decoded and mixed with speech priority balance.

10. Transition Architecture (12 GPU Types)

Editor FS provides strict visual symmetry between **Live Preview** in Flutter and **Hardware MP4 Export** on Android across all 12 transition types:

Type Key	Name	Live Preview Implementation (Dart)	Export Engine Implementation (Kotlin)
`fade`	Fade	Cross-fading `Opacity` widgets (clip A: $1-t$, clip B: t)	Bitmap alpha blending with PorterDuff
`dissolve`	Dissolve	Progressive cross-dissolve opacity blending	Direct alpha blending on decoded video frames
`blackFade`	Fade Black	Fade out to black container, then fade in clip B	Color matrix multiplication through black
`whiteFade`	Fade White	Flash fade through pure white canvas overlay	Color matrix flash blending through white
`slideLeft`	Slide Left	`Transform.translate` offset from 0 to $-W$	Canvas 2D translation matrix moving left
`slideRight`	Slide Right	`Transform.translate` offset from 0 to $+W$	Canvas 2D translation matrix moving right
`slideUp`	Slide Up	`Transform.translate` offset from 0 to $-H$	Canvas 2D translation matrix moving up
`slideDown`	Slide Down	`Transform.translate` offset from 0 to $+H$	Canvas 2D translation matrix moving down
`wipeLeft`	Wipe Left	`ClipRect` revealing clip B right-to-left	Bitmap canvas clipping rectangle right-to-left
`wipeRight`	Wipe Right	`ClipRect` revealing clip B left-to-right	Bitmap canvas clipping rectangle left-to-right
`zoomIn`	Zoom In	`Transform.scale` expanding clip B from 0.5 to 1.0	Scale matrix centered at $(W/2, H/2)$ expanding
`zoomOut`	Zoom Out	`Transform.scale` contracting clip A from 1.0 to 0.5	Scale matrix centered at $(W/2, H/2)$ shrinking

Transition Mathematical Normalization

Both preview and export calculate the normalized transition progress parameter:

```
t = clamp((playheadTime - transitionStartTime) / transitionDuration, 0.0, 1.0)
```

Ensures that frame F rendered in preview at timestamp T matches frame F rendered by the hardware encoder at timestamp T.

11. Hardware Export Architecture

The export pipeline runs entirely in native Android C++/Java APIs using hardware video acceleration:

HARDWARE CONSTRAINTS & PIPELINE GUARANTEES:

- 16-Pixel Dimension Clamping: Width and Height are clamped to multiples of 16 (e.g. 1080x1920) for hardware encoder compliance.
- Scoped Storage Compliance: Uses Android 10+ MediaStore with IS_PENDING=1 during encoding; flipped to 0 on final muxer release.
- Real-time Progress Broadcast: Native engine computes (encodedFrames / totalFrames) and pulses progress events back to Flutter.

12. Platform Channel Map

Complete matrix of all inter-process communication channels crossing the Flutter/Android boundary:

Channel Identifier	Type	Direction	Flutter Facade	Native Android Handler	Methods / Events Handled
`com.example.capcut_video_editor/video_player`	Method	Flutter → Android	`VideoPlaybackService`	`MainActivity.kt` → `VideoPlaybackEngine`	`initialize`, `play`, `pause`, `seekTo`, `setSpeed`, `setVolume`, `exportVideo`
`com.example.capcut_video_editor/video_player_event`	Event	Android → Flutter	`VideoPlaybackService`	`MainActivity.kt` (16ms Ticker)	`onInitialized`, `onPositionUpdate`, `onVideoCompleted`, `onError`
`com.example.capcut_video_editor/file_picker`	Method	Flutter → Android	`FilePickerService`	`MainActivity.kt` → `MediaAssetEngine`	`pickVideo`, `pickAudio`, `pickImage`, `getVideoDuration`, `getVideoThumbnail`, `extractAudioFromVideo`
`com.example.capcut_video_editor/audio_player`	Method	Flutter → Android	`AudioPlayerService`	`MainActivity.kt` → `AudioEngine`	`loadAudio`, `play`, `pause`, `seekTo`, `setVolume`, `startRecording`, `stopRecording`
`com.example.capcut_video_editor/audio_player_event`	Event	Android → Flutter	`AudioPlayerService`	`MainActivity.kt` → `AudioEngine`	`onAudioInitialized`, `onAudioPositionUpdate`, `onAudioCompleted`
`com.example.capcut_video_editor/tts`	Method	Flutter → Android	`TtsService`	`MainActivity.kt` → `TextToSpeech`	`synthesizeToFile` (returns audio file path)

13. Core Data Flow Maps (20 Flows)

User Action / Flow	Execution Sequence (Flutter Layer)	Execution Sequence (Native Layer)
1. App Startup	`main()` → setup providers → `HomeScreen` loads drafts	`MainActivity.configureFlutterEngine()` binds all channels
2. Create Project	User taps "+ New Project" → opens media picker	No-op until media selection confirmed
3. Import Video	`FilePickerService.pickVideo()`	SAF Intent → `MediaAssetEngine` extracts duration & thumbnail
4. Import Audio	`FilePickerService.pickAudio()`	SAF Intent → returns audio URI & duration to ViewModel
5. Add to Timeline	Appends `MediaAsset` to `Project.tracks[0]`	Calls `initialize(uri)` → returns `textureId`
6. Play	`EditorViewModel.play()` sets `isPlaying=true`	`VideoPlaybackEngine.play()` + starts 16ms ticker
7. Pause	`EditorViewModel.pause()` sets `isPlaying=false`	`VideoPlaybackEngine.pause()` + stops 16ms ticker
8. Seek	User drags scrubber → `seekTo(sec)`	`VideoPlaybackEngine.seekTo(msec)`
9. Playhead Sync	`_handleVideoPositionEvent(sec)` updates playhead	Ticker queries `MediaPlayer.currentPosition` every 16ms
10. Split Clip	`splitCurrentClip()` splits `MediaAsset` at playhead	Pure domain model operation; preserves file references
11. Trim Clip	User drags clip edges → adjusts `startTime/endTime`	Recalculates timeline bounds & clamps playhead
12. Delete Clip	Removes clip from primary track	Removes adjacent transitions; ripples remaining clips
13. Ripple Delete	Deletes clip & shifts all subsequent clips left	Recalculates total project duration; syncs playhead
14. Extract Audio	`extractAudio()` → `extractAudioFromVideo()`	`MediaExtractor` + `MediaMuxer` extracts M4A in cache
15. Add SFX	Selects sound from `AssetLibraryService`	Loads into `AudioEngine.SoundPool`
16. Add Transition	`addTransition(type, duration)` between clips A & B	Validates bounds ≤ half minimum clip length
17. Preview Transition	`VideoPreviewSection._buildTransitionCanvas()`	Computes progress t; renders 12-transition transform matrix
18. Export Project	`ExportScreen` → `exportVideo(config)`	`VideoExportEngine` runs MediaCodec, GLES20, and Muxer
19. Save Draft	`ProjectStorageService.saveProject(project)`	Encodes project graph to JSON; writes to local app sandbox
20. Load Draft	`ProjectStorageService.loadProject(id)`	Parses JSON; reinstantiates models; loads native textures

14. Class Relationship Map

```
┌────────────────────────────────┐
│          EditorScreen          │
└────────────────┬───────────────┘
                 │ listens to / dispatches
                 ▼
┌─────────────────────────────────────────────────────────────────┐
│                        EditorViewModel                          │
│  - project: Project                                             │
│  - playheadPosition: double                                     │
│  - isPlaying: bool                                              │
│  - undoStack: List<Project>                                     │
│  - redoStack: List<Project>                                     │
└───┬───────────────────────────────┬─────────────────────────────┘
    │ owns & serializes             │ delegates to
    ▼                               ▼
┌───────────────────────────┐   ┌─────────────────────────────────────────┐
│        Project            │   │ Platform Services:                      │
│  - id: String             │   │  • VideoPlaybackService (Method/Event)  │
│  - title: String          │   │  • AudioPlayerService (Method/Event)    │
│  - tracks: List<Track>    │   │  • FilePickerService (SAF/Extraction)   │
│  - transitions: List<...> │   │  • ProjectStorageService (JSON Disk)    │
└───┬───────────────────────┘   └─────────────────────────────────────────┘
    │ contains items
    ├────────────────────────┬───────────────────────┬─────────────────────┐
    ▼                        ▼                       ▼                     ▼
┌──────────────────┐  ┌──────────────────┐  ┌─────────────────┐  ┌──────────────────┐
│   MediaAsset     │  │   AudioTrack     │  │   TextOverlay   │  │ TransitionEffect │
│ (Video/Overlay)  │  │  (BGM/SFX/M4A)   │  │  (Style/Coords) │  │  (12 GPU Types)  │
└──────────────────┘  └──────────────────┘  └─────────────────┘  └──────────────────┘
```

15. Persistence & Data Model Map

Project graphs are fully persisted to disk as JSON documents in `$APP_SANDBOX/projects/{id}.json`.

Domain Model	Persisted?	Key JSON Fields Stored	Fidelity Guarantee
`Project`	YES	`id`, `title`, `aspectRatio`, `tracks`, `transitions`, `updatedAt`	Full graph recreated on load.
`MediaAsset`	YES	`id`, `localPath`, `duration`, `startTime`, `endTime`, `speed`, `volume`	File existence verified on reload.
`AudioTrack`	YES	`id`, `localPath`, `timelineStart`, `clipDuration`, `volume`, `type`	Timing and volume preserved.
`TransitionEffect`	YES	`id`, `type` (string), `duration`, `fromAssetId`, `toAssetId`	12 types deserialize losslessly.
`TextOverlay`	YES	`text`, `fontFamily`, `fontSize`, `colorHex`, `x`, `y`, `scale`, `duration`	Styling & coordinates restored.
`StickerOverlay`	YES	`stickerId`, `assetPath`, `x`, `y`, `scale`, `rotation`, `duration`	Position & rotation restored.

16. Feature → File Reference Map

Feature Area	Primary Flutter File	Primary Method / Class	Native Android File	Related Model
Video Playback	`video_playback_service.dart`	`play()` / `pause()`	`VideoPlaybackEngine.kt`	`MediaAsset`
Authoritative Clock	`editor_view_model.dart`	`_handleVideoPositionEvent()`	`MainActivity.kt` (16ms)	`Project`
Timeline Scrubber	`timeline_widget.dart`	`_handleHorizontalDragUpdate()`	None (Flutter Gesture)	`TimelineTrack`
Clip Split	`editor_view_model.dart`	`splitCurrentClip()`	None (Domain Logic)	`MediaAsset`
Clip Trim	`editor_view_model.dart`	`trimClip(assetId, start, end)`	None (Domain Logic)	`MediaAsset`
Ripple Delete	`editor_view_model.dart`	`deleteClip()`	None (Domain Logic)	`Project`
12 Transitions	`video_preview_section.dart`	`_buildTransitionCanvas()`	`VideoExportEngine.kt`	`TransitionEffect`
Hardware Export	`export_screen.dart`	`exportVideo(config)`	`VideoExportEngine.kt`	`ExportPreset`
Audio Extraction	`file_picker_service.dart`	`extractAudioFromVideo()`	`MediaAssetEngine.kt`	`AudioTrack`
SFX Library	`audio_bottom_sheet.dart`	`AssetLibraryService`	`AudioEngine.kt`	`AudioTrack`
Voiceover Record	`audio_bottom_sheet.dart`	`startRecording()`	`AudioEngine.kt`	`AudioTrack`
Text-to-Speech	`tts_service.dart`	`synthesizeToFile()`	`MainActivity.kt` (TTS)	`AudioTrack`
Text Overlays	`text_bottom_sheet.dart`	`addTextOverlay()`	None (Flutter Canvas)	`TextOverlay`
Sticker Overlays	`sticker_bottom_sheet.dart`	`addSticker()`	None (Flutter Canvas)	`StickerOverlay`
Draft Persistence	`project_storage_service.dart`	`saveProject()` / `loadProject()`	None (Dart IO JSON)	`Project`

17. “Where Do I Change This?” Developer Table

Developer Task	Exact File & Symbol	Notes & Architectural Rule
Add a New Transition	1. `lib/models/transition_effect.dart` (Enum) 2. `video_preview_section.dart` (`_buildTransitionCanvas`) 3. `VideoExportEngine.kt` (`compositeTransition`)	Must maintain preview/export mathematical symmetry.
Fix Playhead / Clock Drift	1. `MainActivity.kt` (`startVideoPositionUpdates`) 2. `editor_view_model.dart` (`_handleVideoPositionEvent`)	Ensure no Flutter software timer competes with native clock.
Change Export Resolution / Bitrate	`android/` ... `/VideoExportEngine.kt` (`setupEncoder`)	Width & height must remain multiples of 16 for H.264.
Adjust Timeline Zoom / Scrubber	`lib/presentation/widgets/timeline_widget.dart`	Updates pixelsPerSecond multiplier and drag physics.
Add Built-In SFX / Music	`lib/services/asset_library_service.dart`	Register bundled asset paths in registry list.
Change Draft Save Format	1. `lib/models/project.dart` (`toJson` / `fromJson`) 2. `lib/services/project_storage_service.dart`	Maintain backward compatibility for existing project drafts.
Add New Platform Channel	1. `MainActivity.kt` (Channel registration & dispatcher) 2. New service in `lib/services/`	Register binary messenger channel string uniformly.

18. Architectural Boundaries

The codebase enforces seven strict architectural boundaries that must not be violated:

1. Flutter vs. Android Native Boundary
Flutter handles presentation, gestures, state graphs, and user controls. Native Android handles hardware video decoding, SurfaceTextures, OpenGL rendering, and hardware MediaCodec MP4 encoding. No decoded raw video frames cross into Dart memory.

2. Clock Ownership Boundary
The native `MediaPlayer` via `MainActivity`'s 16ms periodic ticker is the sole authoritative clock. The Flutter `EditorViewModel` listens passively and never increments the playhead using a software timer during active video playback.

3. Transition Symmetry Boundary
Every transition effect must have two implementations that share identical normalization logic ($t \in [0.0, 1.0]$): a Dart implementation in `video_preview_section.dart` and a Kotlin OpenGL/Canvas implementation in `VideoExportEngine.kt`.

4. Persistence Isolation Boundary
The `Project` graph serializes strictly to JSON. It holds only file paths and timing metrics, never raw bytes or UI state (such as scroll offset or zoom level).

5. Zero Third-Party Media Dependencies
The project deliberately avoids external media libraries (e.g. `flutter_ffmpeg`, `audioplayers`, `video_player`) to maintain direct control over hardware frame timing, EGL surfaces, and export pipelines.

19. Final Architecture Summary Sheet

Condensed Quick-Reference Card for Ongoing Development

EDITOR FS ARCHITECTURE AT A GLANCE

```
Presentation  → EditorScreen (VideoPreviewSection, TimelineWidget, BottomToolbar, Tool Sheets)
State         → EditorViewModel (Authoritative Clock, Multi-Track State, Undo/Redo Stacks)
Domain Models → Project, MediaAsset, AudioTrack, TextOverlay, StickerOverlay, TransitionEffect
Services      → VideoPlaybackService, AudioPlayerService, FilePickerService, TtsService, Storage
Channels      → video_player (Method/Event), file_picker, audio_player (Method/Event), tts
Native Android→ MainActivity (16ms Ticker), VideoPlaybackEngine, VideoExportEngine, AudioEngine
Hardware/IO   → MediaPlayer, MediaCodec H.264/AAC, EGL14/GLES20, MediaMuxer, Scoped MediaStore
```

Feature Component	Flutter Location	Native Location	Key Function
Video Decoding / Preview	`video_preview_section.dart`	`VideoPlaybackEngine.kt`	`initialize`, `Texture`
Playhead Synchronization	`editor_view_model.dart`	`MainActivity.kt`	`_handleVideoPositionEvent`
Timeline Editing / Scrubber	`timeline_widget.dart`	None	`_handleHorizontalDragUpdate`
12 GPU Transitions	`video_preview_section.dart`	`VideoExportEngine.kt`	`_buildTransitionCanvas`
Hardware MP4 Export	`export_screen.dart`	`VideoExportEngine.kt`	`exportVideo`, `MediaCodec`
Audio Mixing & Demux	`audio_bottom_sheet.dart`	`AudioEngine.kt` / `MediaAssetEngine`	`extractAudioFromVideo`
Draft Save / Load	`project_storage_service.dart`	None	`saveProject`, `loadProject`

— END OF ARCHITECTURE & CODE MAP MANUAL —
Editor FS • com.example.capcut_video_editor • Version 2.4.8